

IT PROFESSIONAL PORTFOLIO

Cloud Engineering and Security Projects

PROJECT 33

Keep Private Subnets Private with VPC Endpoints

Amazon VPC | Interface Endpoints | Systems Manager | PrivateLink | EC2

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ENVIRONMENT	Personal AWS Lab
VERSION	v1.0.0
RELEASE DATE	July 22, 2026
STATUS	Complete

Executive Summary

Logged into **TestAccount1**, created a new **SLAW** CloudFormation stack from the provided template URL, verified the private EC2 instance, and created three **interface VPC endpoints** required for Session Manager connectivity without NAT Gateway Internet egress. The endpoints were named **ssm-1**, **ssm-2**, and **ssm-3**, and mapped to **ssm**, **ssmmessages**, and **ec2messages**. Each endpoint was placed in **slaw-private-1** and associated with the **SLAW-SLAWSecurityGroup** security group. After validating Session Manager connectivity, the endpoints and CloudFormation stack were deleted for cost control.

SECURITY OUTCOME

This lab demonstrates private control-plane access. Session Manager worked without NAT Gateway Internet egress by using interface VPC endpoints for the AWS Systems Manager API paths.

Business Problem

In the previous EC2 lab, the instance lived in a private subnet but still depended on NAT Gateway outbound Internet access to reach Systems Manager service endpoints. For stronger isolation, private subnets should be able to access selected AWS services without broad Internet routing. Interface VPC endpoints provide that private service path by placing Elastic Network Interfaces in the subnet and resolving AWS service DNS names to private endpoint addresses.

Objectives

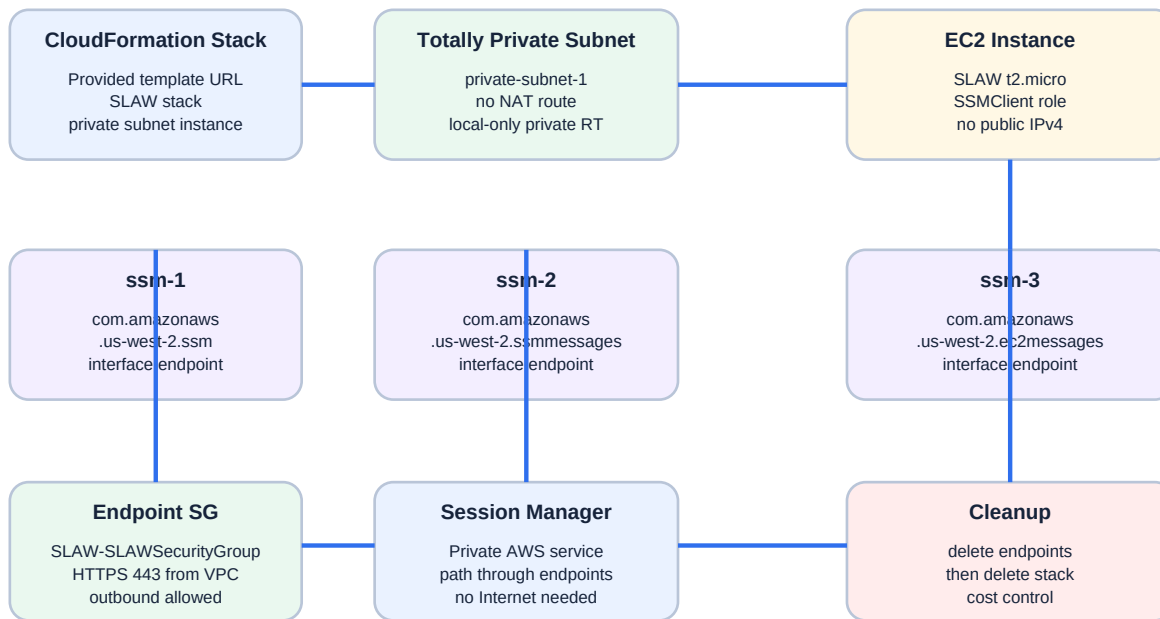
- Deploy the provided CloudFormation template in TestAccount1 to create the SLAW private-subnet test environment.
- Verify the EC2 instance is running before testing Session Manager access.
- Create three interface VPC endpoints required by the SSM Agent: ssm, ssmmessages, and ec2messages.
- Name the endpoints ssm-1, ssm-2, and ssm-3.
- Place each endpoint in private subnet slaw-private-1.
- Attach the SLAW-SLAWSecurityGroup security group to each endpoint.
- Wait for endpoint status to become Available.
- Reboot the EC2 instance when Session Manager was not ready so DNS/cache state refreshed.
- Confirm Session Manager connection to the instance.
- Delete the VPC endpoints and CloudFormation stack to control lab cost.

Environment

Cloud Provider	Amazon Web Services (AWS)

Primary Services	Amazon VPC, VPC Endpoints, AWS Systems Manager Session Manager, Amazon EC2, AWS CloudFormation
Account Context	TestAccount1 / AdministratorAccess
Stack Name	SLAW
Template Source	Provided CloudFormation template URL from the lab
VPC Pattern	Private subnet instance without NAT Gateway private route to 0.0.0.0/0
EC2 Instance	SLAW, t2.micro, running in us-west-2a
Endpoint 1	ssm-1 -> com.amazonaws.us-west-2.ssm
Endpoint 2	ssm-2 -> com.amazonaws.us-west-2.ssmmessages
Endpoint 3	ssm-3 -> com.amazonaws.us-west-2.ec2messages
Endpoint Type	Interface
Endpoint Subnet	slaw-private-1
Endpoint Security Group	SLAW-SLAWSecurityGroup
Validation	Session Manager browser terminal connected
Cleanup	Deleted VPC endpoints, then deleted CloudFormation stack

Architecture



Three interface endpoints replace NAT Internet egress for Session Manager control-plane access.

Figure 1 - Session Manager connectivity through private interface VPC endpoints.

Implementation Summary

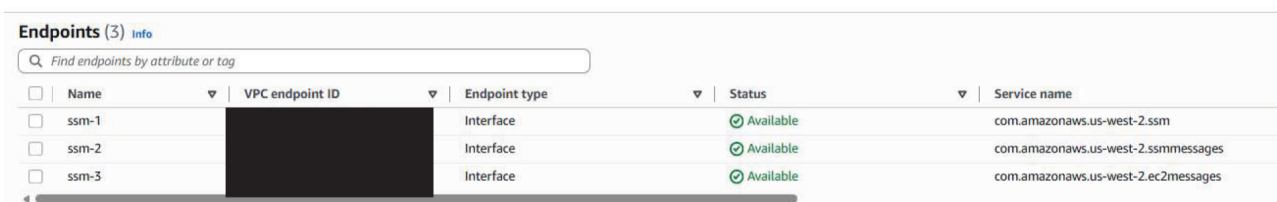
1. Signed into TestAccount1 with AdministratorAccess.
2. Created the SLAW CloudFormation stack using the provided template URL.
3. Verified that the stack created the private EC2 instance and supporting VPC resources.
4. Opened VPC Endpoints and created ssm-1 for com.amazonaws.us-west-2.ssm.
5. Created ssm-2 for com.amazonaws.us-west-2.ssmmessages.
6. Created ssm-3 for com.amazonaws.us-west-2.ec2messages.
7. Selected interface endpoint type for all three endpoint services.
8. Assigned each endpoint to slaw-private-1.
9. Attached SLAW-SLAWSecurityGroup to each endpoint so HTTPS-based AWS API traffic could reach the endpoint network interfaces.
10. Waited for all three endpoints to show Available.
11. Opened EC2 and verified the SLAW instance was running and passing status checks.
12. Rebooted the instance because Session Manager was initially not ready, which refreshed the agent/DNS path.
13. Connected to the instance through Session Manager and confirmed terminal access.
14. Deleted the three VPC endpoints and then deleted the CloudFormation stack for cost control.

Endpoint Service Map

Endpoint Name	Service Name	Description
ssm-1	com.amazonaws.us-west-2.ssm	Primary Systems Manager API service endpoint.
ssm-2	com.amazonaws.us-west-2.ssmmessages	Session Manager message channel service used by the SSM Agent.
ssm-3	com.amazonaws.us-west-2.ec2messages	EC2 message channel used by Systems Manager managed instances.
Endpoint subnet	slaw-private-1	Places endpoint ENIs inside the private subnet for private service access.
Endpoint security group	SLAW-SLAWSecurityGroup	Allows HTTPS API traffic to the endpoint interfaces.

Evidence

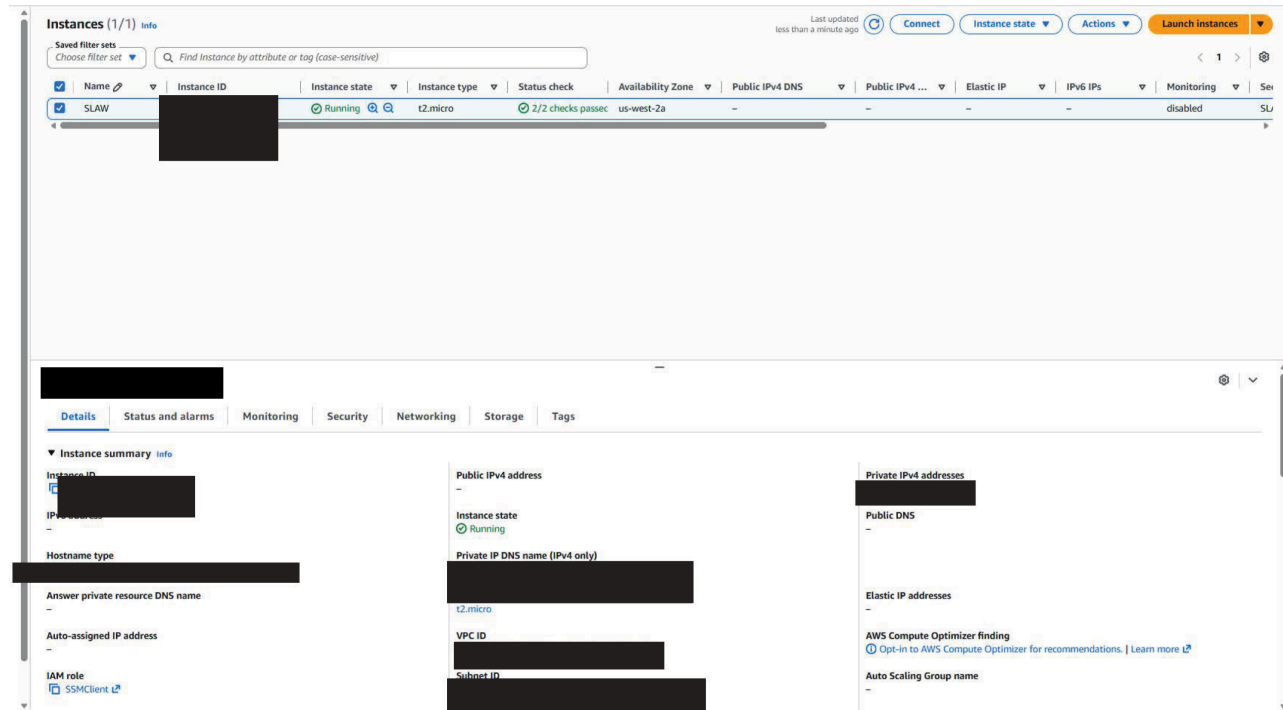
Figure 2 - Three Interface VPC Endpoints Available



The screenshot shows the AWS VPC console 'Endpoints (3)' page. It features a search bar and a table with columns for Name, VPC endpoint ID, Endpoint type, Status, and Service name. Three endpoints are listed: ssm-1, ssm-2, and ssm-3. All three are of type 'Interface' and have a status of 'Available'. The service names are com.amazonaws.us-west-2.ssm, com.amazonaws.us-west-2.ssmmessages, and com.amazonaws.us-west-2.ec2messages. The VPC endpoint IDs are redacted with black boxes.

☐	Name	VPC endpoint ID	Endpoint type	Status	Service name
☐	ssm-1		Interface	Available	com.amazonaws.us-west-2.ssm
☐	ssm-2		Interface	Available	com.amazonaws.us-west-2.ssmmessages
☐	ssm-3		Interface	Available	com.amazonaws.us-west-2.ec2messages

The VPC endpoint evidence shows three endpoints named ssm-1, ssm-2, and ssm-3. All are interface endpoints in Available state. The service names shown are com.amazonaws.us-west-2.ssm, com.amazonaws.us-west-2.ssmmessages, and com.amazonaws.us-west-2.ec2messages. Endpoint IDs are redacted.

Figure 3 - EC2 Instance Running in Private Configuration

The EC2 evidence shows the SLAW instance running as t2.micro in us-west-2a with 2/2 status checks passed. Public IPv4 address, public IPv4 DNS, Elastic IP, and IPv6 IPs are blank, and the details pane shows the SSMClient IAM role. Sensitive identifiers were redacted.

Figure 4 - Session Manager Connection Confirmed

The connection evidence shows an active browser-based Session Manager terminal with a shell prompt and a Terminate session button, confirming the private endpoint path supported Session Manager access.

Security Analysis

This project improves the previous private-subnet pattern by removing the need for NAT Gateway Internet egress for Session Manager access.

VPC endpoints reduce broad outbound exposure. The instance can reach the specific AWS services needed for management without needing a route to the public Internet.

The design depends on DNS behavior. With private DNS enabled, calls to Systems Manager service names resolve to private endpoint addresses inside the VPC.

The three endpoints are required because Session Manager uses multiple service APIs. Creating only the ssm endpoint is not enough for reliable Session Manager operation.

Associating SLAW-SLAWSecurityGroup with the endpoint ENIs is important because interface endpoints are network interfaces and require security group control.

A reboot can resolve temporary readiness problems because software agents and DNS caches may need to refresh before the new endpoint path is used.

Cost and Cleanup Analysis

Interface endpoints are billed hourly, and this lab uses three of them. The lab explicitly treats endpoint deletion as a required cleanup step.

The correct cleanup order is to delete the VPC endpoints first, then delete the CloudFormation stack. The stack can remove the resources it created, but the manually created endpoints must be removed first.

Deleting the CloudFormation stack after endpoint deletion also removes the EC2 instance and VPC resources created by the template.

Lessons Learned

A subnet can be private from the Internet but still access AWS services through VPC endpoints.

Interface endpoints create ENIs in subnets and require security groups.

Some AWS services require multiple VPC endpoints because their agents or clients talk to multiple API endpoints.

Private DNS is what makes normal service names resolve to the endpoint instead of the public AWS API endpoint.

Session Manager readiness may lag after endpoint creation; waiting or rebooting can refresh the endpoint/DNS path.

Production Improvements

- Deploy endpoints in multiple Availability Zones for resilience.
- Use VPC endpoints for Systems Manager, EC2Messages, SSMMessages, S3, CloudWatch Logs, and other required private services.

- Restrict endpoint security groups to the minimum required source security groups or CIDR ranges.
- Enable Session Manager logging to CloudWatch Logs or S3.
- Apply endpoint policies where supported to limit allowed API actions and principals.
- Manage endpoints with CloudFormation or Terraform instead of manual console creation.
- Monitor endpoint creation, deletion, and policy changes with CloudTrail and EventBridge.
- Use AWS Config/Security Hub controls to detect public IP assignment and unmanaged Internet egress.

Skills Demonstrated

VPC Endpoints	Session Manager	CloudFormation stack lifecycle
Interface endpoints	Private DNS	SSM Agent troubleshooting
Security groups for ENIs	Private EC2 administration	Endpoint cleanup

Resume Bullet

RESUME

Implemented private Systems Manager access for an EC2 instance by creating interface VPC endpoints for ssm, ssmmessages, and ec2messages, associating each endpoint with a private subnet and security group, validating Session Manager connectivity, and deleting endpoints and stack resources for cost control.

STAR Interview Story

Situation: The previous EC2 lab used a private instance, but Session Manager still depended on NAT Gateway outbound Internet access.

Task: Replace that Internet-dependent path with private AWS service connectivity while still allowing administrative access to the instance.

Action: I deployed the SLAW stack, created three interface VPC endpoints for ssm, ssmmessages, and ec2messages in slaw-private-1, attached SLAW-SLAWSecurityGroup, verified endpoint availability and instance health, rebooted the instance when Session Manager was not ready, confirmed terminal access, and deleted endpoints and stack resources.

Result: The lab proved that a private EC2 instance can be administered through Session Manager without inbound or outbound Internet access, while maintaining cost-control cleanup discipline.

Version History

v1.0.0	July 22, 2026	Initial documented implementation. Created SLAW stack from provided template, created ssm-1, ssm-2, and ssm-3 interface VPC endpoints, associated each with slaw-private-1 and SLAW-SLAWSecurityGroup, verified EC2 and Session Manager access, rebooted instance during troubleshooting, deleted endpoints and stack for cost control, and embedded provided evidence using portfolio documentation standard v1.0.1.

References

- Cloud Security Lab a Week (S.L.A.W.) - Keep Private Subnets Private with VPC Endpoints.
- User-provided VPC endpoints evidence screenshot.
- User-provided EC2 instance validation screenshot.
- User-provided Session Manager connection screenshot.
- Project 32 - Launch a Private Amazon Linux EC2 Instance with Session Manager.